

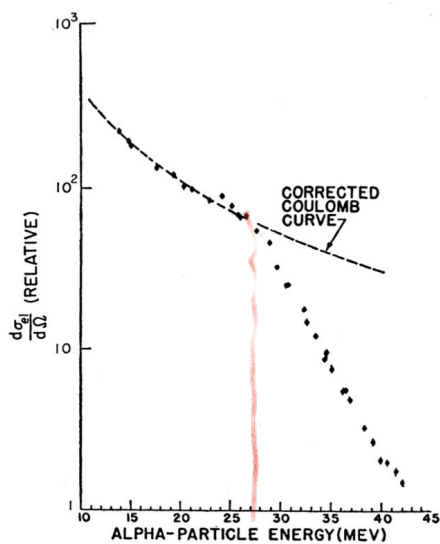


FIG. 5. Differential cross section for the elastic scattering of alpha particles by Pb at 60° as a function of the alpha-particle energy.

$$\frac{d\sigma}{d\Omega} = \underbrace{\frac{(\alpha Z z_1)^2}{q^4}}_{\text{Rutherford}} \underbrace{\frac{|\vec{p}_{\text{out}}|}{|\vec{p}_{\text{in}}|}}_{\text{Relativ}} E'^2 (1 - \beta^2 \sin^2 \frac{\theta}{2})$$

Mott

point-like probe with $S = \frac{1}{2}$

point-like target / no spin

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Mott}} = \left(\frac{d\sigma}{d\Omega}\right)_{\text{Ruth}} \frac{E'}{E} E' (1 - \beta^2 \sin^2 \frac{\theta}{2})$$



Nuclear radii: $R = r_0 A^{1/3}$ $r_0 = 1.1 \text{ fm}$

$$\Delta p \cdot \Delta x \sim 1 \Rightarrow \Delta p \approx \frac{1}{\Delta x} \quad \frac{1}{1 \text{ fm}} = 200 \text{ MeV}$$

Nucleus as a Fermi gas of nucleons (p, n)

$$m_N \sim 1 \text{ GeV} \quad \frac{p}{m} = 0.2 \ll 1$$

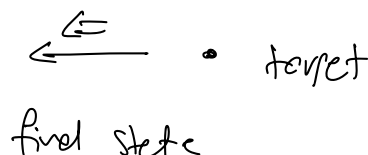
$$E = \sqrt{p^2 + m^2} \approx m + \frac{p^2}{2m} = m + K$$

$$K_N = \frac{p^2}{2m_N} = \frac{4 \times 10^4 \text{ MeV}^2}{2 \times 10^3 \text{ MeV}} = 20 \text{ MeV}$$

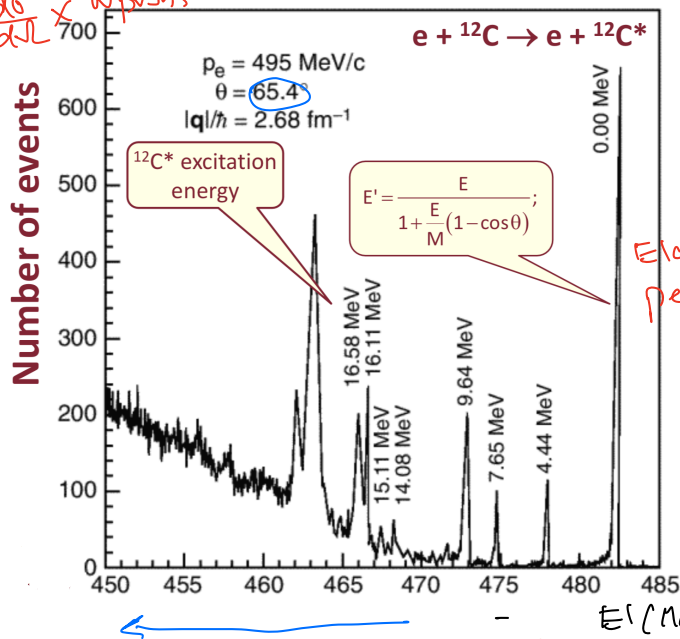
α particle is interacting with ingredients with $E \sim 20 \text{ MeV}$

At high energy limit $\beta \rightarrow 1$

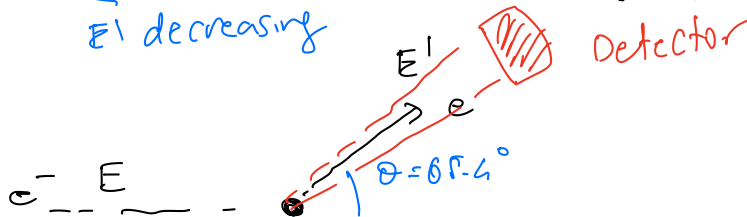
$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{Mott}} = \left(\frac{d\sigma}{d\Omega}\right)_{\text{Ruth}} \underbrace{\frac{E'}{E}}_{=1} E'^2 \cos^2 \frac{\theta}{2}$$



$\int \frac{d\sigma}{d\Omega} \propto \text{approx}$



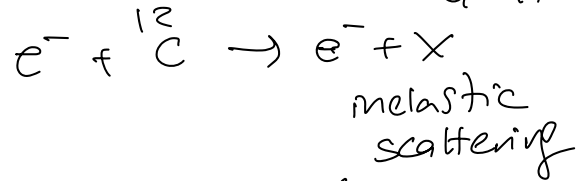
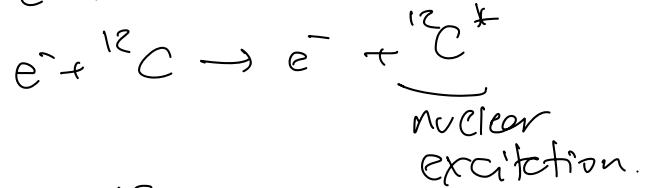
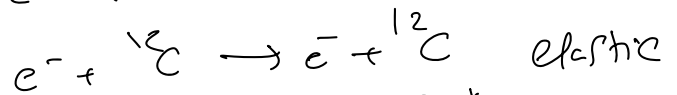
E' decreasing



print like e^- against ${}^{12}\text{C}$

$$p_e = 495 \text{ MeV} \gg m_e$$

$$E_e = 495 \text{ MeV}$$



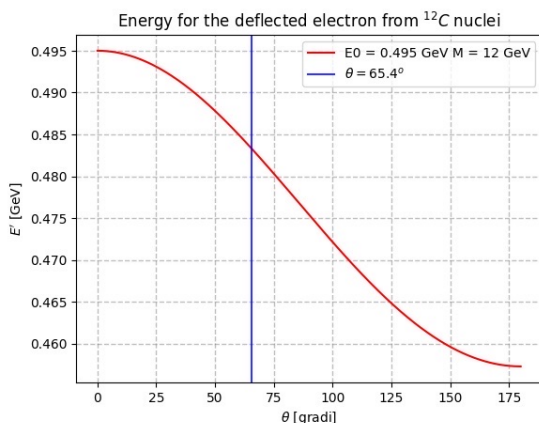
$$\# \text{event} \propto \int_{\theta-\epsilon}^{\theta+\epsilon} \frac{d\sigma}{d\Omega} d\Omega$$

$$\frac{E'}{E} = \frac{1}{1 + \frac{E}{M}(1 - \cos\theta)} = \frac{1}{1 + \frac{2E}{M} \sin^2 \frac{\theta}{2}}$$

$$\frac{E'}{E} \approx \frac{0.5 \text{ GeV}}{12 \text{ GeV}}$$

$$E' = E \frac{1}{1 + 0.03} \approx (1 - 0.03) E$$

$$= E' \times 0.97 = 480 \text{ MeV}$$



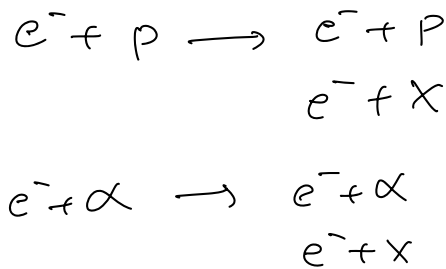
$$\frac{d\sigma}{d\Omega} = \frac{(\alpha Z z)^2}{q^4}$$

$$E'^2 (1 - \beta^2 \sin^2 \frac{\theta}{2})$$

$$\frac{1}{1 + \frac{2E}{M} \sin^2 \frac{\theta}{2}}$$

$$\int_{\theta-\epsilon}^{\theta+\epsilon} \frac{d\sigma}{d\Omega} d\varphi d\cos\theta$$

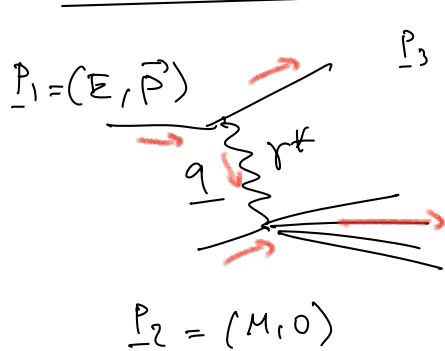
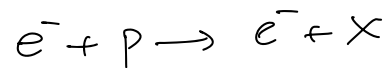
next step to study structure of nucleons (p.n).



Experiments in 1960s
SLAC (Stanford)
DESY (Hamburg)

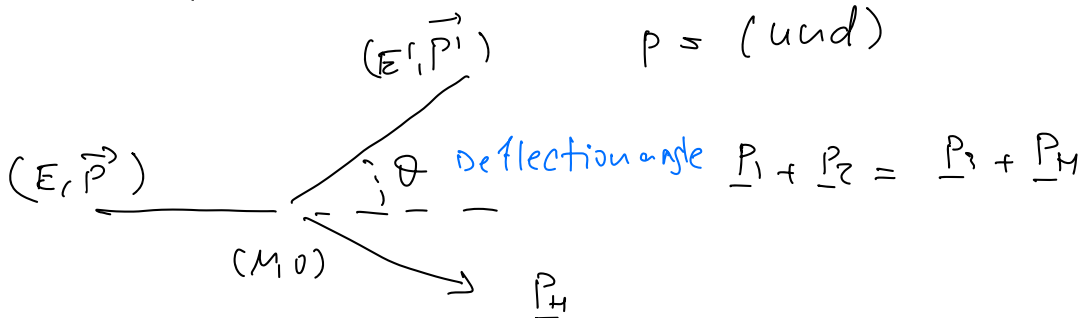
Increase energy from 0.5 GeV $\rightarrow$ 20 GeV

Elastic Scattering Kinematics



$$\underline{p}_4 = \underline{p}_2 + \underline{q}$$

$$\underline{p}_4 = \sum_i \underline{p}_{xi} = \underline{p}_H \quad \text{Hadronic system particles in final states}$$



$$p = (uud)$$

$$\underline{p}_1 + \underline{p}_2 = \underline{p}_3 + \underline{p}_4$$

$$\underline{q} = \underline{p}_1 - \underline{p}_3 = (E - E', \vec{p} - \vec{p}')$$

4-mom. of virtual photon

$$q^2 = (\underline{p}_1 - \underline{p}_3)^2 = \underline{p}_1^2 + \underline{p}_3^2 - 2 \underline{p}_1 \cdot \underline{p}_3 = 2m_e^2 - 2(E E' - p p' \cos \theta)$$

$$E \gg m_e \Rightarrow E \approx p \quad E' \approx p'$$

$$1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$$

$$q^2 = -2(E E' - E E' \cos \theta) = -2 E E' (1 - \cos \theta) = -4 E E' \sin^2 \frac{\theta}{2}$$

$$q^2 < 0 \quad \forall \theta \Rightarrow \gamma^* \text{ virtual}$$

$$Q^2 := -q^2 \geq 0$$

$$q^2 = t \quad \text{Mandelstam variable}$$

$$(\underline{p}_1 + \underline{p}_2)^2 = s$$

$$\underline{p}_4 = \underline{p}_2 + \underline{q}$$

$$|\underline{p}_4|^2 = \underline{p}_2^2 + \underline{q}^2 + 2 \underline{p}_2 \cdot \underline{q} = M^2 + q^2 + 2M(E - E')$$

$$W^2 = M^2 - Q^2 + 2M(E - E')$$

W^2 : invariant mass of hadronic system.

$Q^2 = -q^2 > 0$: transferred mom. by γ^*

$\nu = E - E'$: energy loss by the probe.

$$W^2 = M^2 - Q^2 + 2M\nu.$$

Experimentally measure $(E', \theta) \Rightarrow$ Compute $\left\{ \begin{array}{l} Q^2 = 4EE' \sin^2 \frac{\theta}{2} \\ \nu = E - E' \end{array} \right.$

E is the beam energy. (Known)

$$Q^2 = 2M\nu + M^2 - W^2$$

Kinematic constraint
in inelastic collisions.

$$e^- + p \rightarrow e^- + X$$

Elastic limit : $X = p \Rightarrow W^2 = M^2$

Bjorken variable $\boxed{Q^2 = 2M\nu}$ Elastic scattering.

$$x = \frac{Q^2}{2M\nu} = \frac{Q^2}{2M(E-E')} \quad (\text{elastic}) \text{ measure } \theta, E' \Rightarrow \text{compute } Q^2, \nu$$

$x = 1 \Rightarrow$ elastic scattering.

$x \neq 1 \Rightarrow$ inelastic.

$$y = \frac{\nu}{E} = \frac{(E-E')}{E} \quad E' \in [0, E] \Rightarrow y \in [0, 1]$$

$$x = \frac{M^2 - W^2 + 2M\nu}{2M\nu} = 1 + \frac{M^2 - W^2}{2M\nu} \quad x \in [0, 1]$$

$$S = (P_1 + P_2)^2 = m_e^2 + M^2 + 2ME \geq M^2 + 2ME$$

$$t = (P_1 - P_3)^2 = q^2 = -Q^2$$

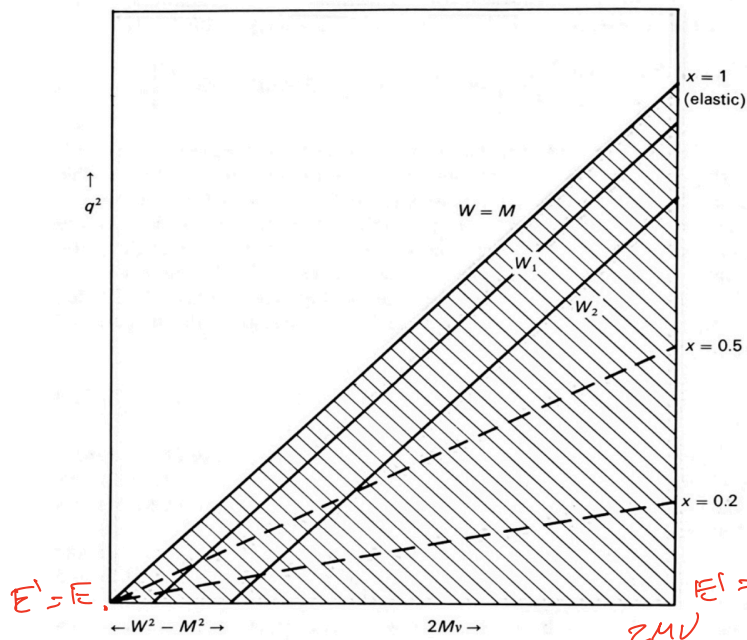


Figure 8.5 (a) Kinematic relations in inelastic lepton-nucleon scattering. The 4-momentum transfer squared, q^2 , is plotted against $2Mv$, where M is the nucleon mass and v is the energy transfer measured in the rest frame of the target. The invariant mass of the final state of hadrons is given by $W^2 = M^2 + 2Mv - q^2$, and the variable $x = q^2/2Mv$. Elastic scattering corresponds to $x = 1$, $W = M$. Contours of fixed W are 45° lines; those of fixed x are radial lines from the origin with slope x .

$$Q^2 = M^2 - w^2 + 2Mv$$

All Kinematic info about hadronic system constrained by

e^- measurements (E', θ)

$$2Mv = E - E'$$

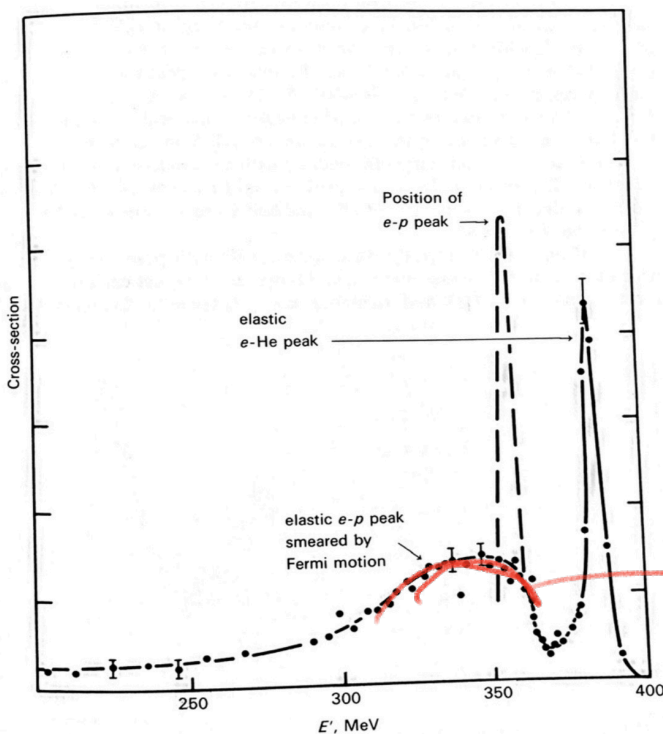
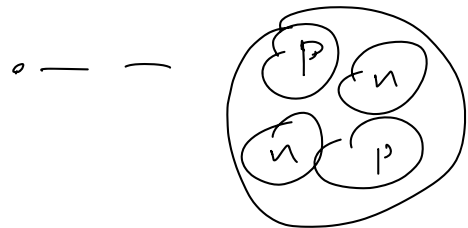


Figure 8.5 (b) Scattering of 400-MeV electrons in helium at 45° , showing a sharp elastic helium peak and a smeared peak due to elastic scattering from individual nucleons. (After Hofstadter 1956.)

$$E_e = 400 \text{ MeV}$$

$$e^- + \alpha \rightarrow e^- + X$$



convolution

$$e^- + p \oplus \text{Gaussian}(\sigma = 25 \text{ MeV})$$

So far we have ignored:

1) finite size & forget spatial distribution of charge in the target

2) spin of the target

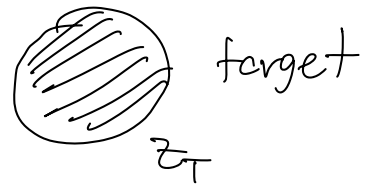
Finite size of target

$$e^- \quad z_p$$

$$\frac{d\sigma}{d\Omega} \propto |M|^2 \rho(\text{phase space})$$

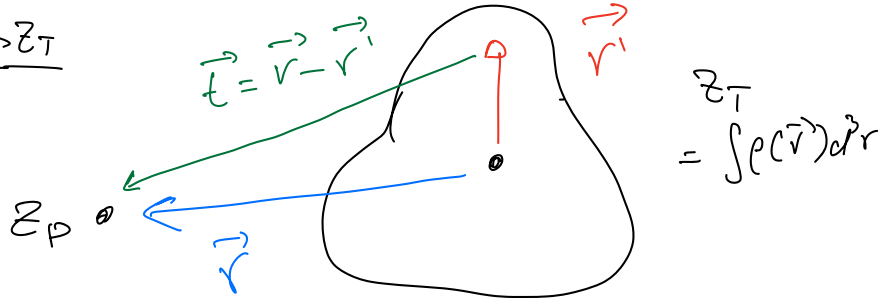
$$M = \langle f | H_I | i \rangle$$

↳ assume spatial distribution.



point like: $H_I = \frac{\alpha z_p z_T}{r}$

$$V(r) = \int d^3 \vec{r}' \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|}$$



point like: $f(\vec{r}) = z_T \frac{\delta(r)}{4\pi}$

$\rho(\vec{r})$ charge distribution.
 ρ : charge density

Extended body with charge distribution

$$\rho(\vec{r}') = z_T f(\vec{r}')$$

$$\int \rho(r) d^3 r = z_T \int f(r) d^3 r = z_T$$

$$\int \rho(\vec{r}') d^3 r' = z_T \int f(\vec{r}') d^3 r' = z_T$$

$$\Rightarrow \int f(r) d^3 r = 1$$

$$\Rightarrow \int f(\vec{r}') d^3 r' = 1$$

Point like object:

$$\int \delta(r) d^3 r = 4\pi \Rightarrow \rho(\vec{r}) = \frac{z_T}{4\pi} \delta(r)$$

$$V(r) = z_T \int d^3 r' \frac{\rho(\vec{r}')}{|\vec{r} - \vec{r}'|}$$

$$\langle f | V | i \rangle = M \sim \int d^3 r \psi_f^* V \psi_i$$

Born approximation: free particles in initial and final state

$$\psi \sim \frac{1}{\sqrt{V}} e^{i \vec{p} \cdot \vec{r}} \quad \text{plane wave.}$$

$$\mu \sim z_T z_P \int d^3\vec{r} \underbrace{e^{-i\vec{P}_f \cdot \vec{r}}}_{\substack{\text{final state} \\ \vec{P}_f}} \underbrace{e^{+i\vec{P}_{in} \cdot \vec{r}}}_{\substack{\text{init state} \\ \vec{P}_{in}}} \int d^3\vec{r}' \underbrace{\frac{f(\vec{r}')}{|\vec{r} - \vec{r}'|}}_{\substack{= \vec{t} \\ \text{Extended body}}}$$

$$e^{+i(\vec{P}_f - \vec{P}_{in}) \cdot \vec{r}} = e^{+i\vec{q} \cdot \vec{r}} \quad \vec{q} = \vec{P}_{in} - \vec{P}_f \quad \text{momentum transfer from prob to target}$$

change of variable: $\vec{t} = \vec{r} - \vec{r}'$

$$\Rightarrow \vec{r} = \vec{t} + \vec{r}' \quad \Rightarrow \quad i\vec{q} \cdot \vec{r} = i\vec{q} \cdot \vec{t} + i\vec{q} \cdot \vec{r}'$$

$$\mu \sim z_T z_P \underbrace{\int d^3\vec{r}' f(\vec{r}') e^{+i\vec{q} \cdot \vec{r}'}}_{F(q^2)} \underbrace{\int d^3\vec{t} e^{+i\vec{q} \cdot \vec{t}} \frac{1}{|\vec{t}|}}_{\frac{1}{q^2}}$$

Form factor in momentum space.

$F(q^2)$ = Fourier transform of $f(\vec{r})$

$$\mu \sim \frac{z_P z_T}{q^2} \underbrace{(F(q^2))}_{\text{Extended body.}}$$

$$\mu \sim \frac{z_P z_T}{q^2} \quad \text{point-like}$$

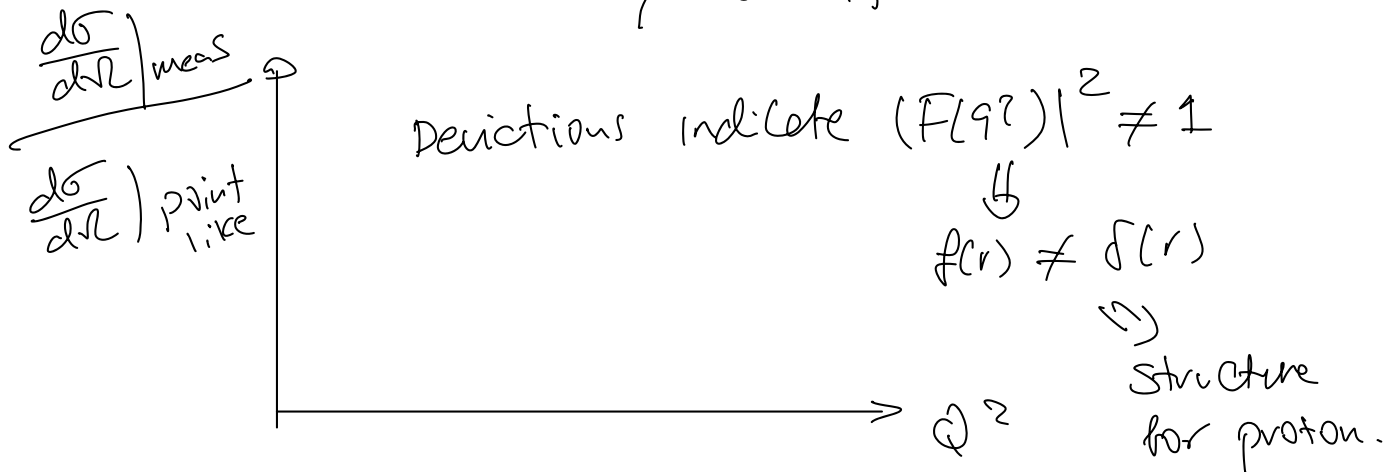
$$\left. \frac{d\sigma}{d\Omega} \right|_{\text{Extended target}} = \left. \frac{d\sigma}{d\Omega} \right|_{\text{point-like}} \times |F(q^2)|^2$$

$$q^2 = -Q^2 = 4EE' \sin^2 \frac{\theta}{2} \quad \text{measured experimentally}$$

Experimental plan:

- $e^- + p \rightarrow e^- + X$ collisions.
- measure Q^2

- study $\frac{d\sigma}{dQ^2} / \text{meas} / \frac{d\sigma}{dQ^2} / \text{point like} \sim F(Q^2)$

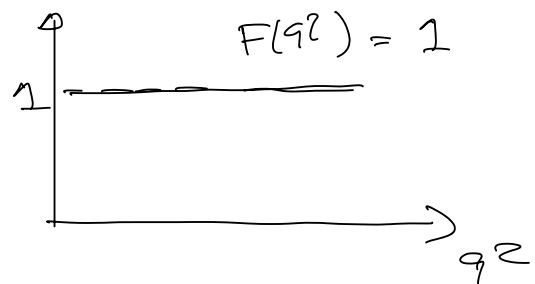


Examples of $f(\vec{r})$:

1) point-like: $f(r) = \frac{\delta(r)}{4\pi}$

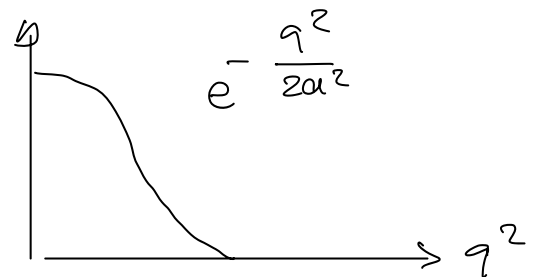
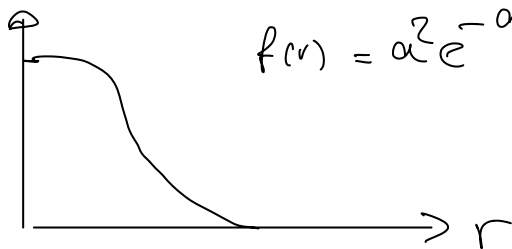


$f(\vec{r}) = f(r, \theta, \phi) \simeq f(r)$



2) Gaussian charge distribution

$f(r) = \alpha^2 e^{-\alpha^2 \frac{r^2}{2}}$



3) Homogeneous ferret

